Time Series Analysis

N.Omkar

What Is Time Series Analysis?

Time series analysis is a specific way of analyzing a sequence of data points collected over time. In TSA, analysts record data points at consistent intervals over a set period rather than just recording the data points intermittently or randomly.

How to Analyze Time Series?

To perform the time series analysis, we have to follow the following steps:

- Collecting the data and cleaning it
- Preparing Visualization with respect to time vs key feature
- Observing the stationarity of the series
- Developing charts to understand its nature.
- Model building – AR, MA, ARMA and ARIMA
- Extracting insights from prediction

Components of Time Series Analysis

Let’s look at the various components of Time Series Analysis:

- Trend: In which there is no fixed interval and any divergence within the given dataset is a continuous timeline. The trend would be Negative or Positive or Null Trend
- Seasonality: In which regular or fixed interval shifts within the dataset in a continuous timeline. Would be bell curve or saw tooth
- Cyclical: In which there is no fixed interval, uncertainty in movement and its pattern
- Irregularity: Unexpected situations/events/scenarios and spikes in a short time span.

	Trend	Seasonality	Cyclical	Irregularity
Time	Fixed Time Interval	Fixed Time Interval	Not Fixed Time Interval	Not Fixed Time Interval
Duration	Long and Short Term	Short Term	Long and Short Term	Regular/Irregular
Visualization				
Nature - I	Gradual	Swings between Up or Down	Repeating Up and Down	Errored or High Fluctuation
Nature – II	Upward/Down Trend	Pattern repeatable	No fixed period	Short and Not repeatable
Prediction Capability	Predictable	Predictable	Challenging	Challenging
Market Model				Highly random/Unforeseen Events – along with white noise.

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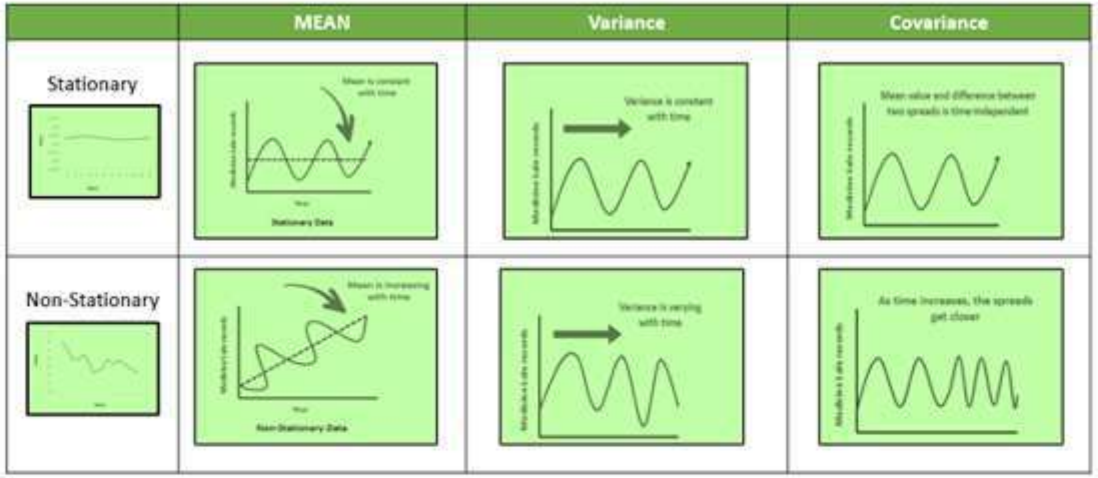
Data Types of Time Series

Let’s discuss the time series’ data types and their influence. While discussing TS data types, there are two major types – stationary and non-stationary.

Stationary: A dataset should follow the below thumb rules without having Trend, Seasonality, Cyclical, and Irregularity components of the time series.

- The mean value of them should be completely constant in the data during the analysis.
- The variance should be constant with respect to the time-frame
- Covariance measures the relationship between two variables.

Non- Stationary: If either the mean-variance or covariance is changing with respect to time, the dataset is called non-stationary.



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Methods to Check Stationarity

- During the TSA model preparation workflow, we must assess whether the dataset is stationary or not. This is done using **Statistical Tests.** There are two tests available to test if the dataset is stationary:
- Augmented Dickey-Fuller (ADF) Test

Augmented Dickey-Fuller (ADF) Test or Unit Root Test

- The ADF test is the most popular statistical test. It is done with the following assumptions:
- Null Hypothesis (H0): Series is non-stationary
- Alternate Hypothesis (HA): Series is stationary
  - p-value >0.05 Fail to reject (H0)
  - p-value <= 0.05 Accept (H1)

Converting Non-Stationary Into Stationary

- **Differencing**

This is a simple transformation of the series into a new time series, which we use to remove the series dependence on time and stabilize the mean of the time series, so trend and seasonality are reduced during this transformation.

Yt= Yt – Yt-1

Yt=Value with time

Moving Average Methodology

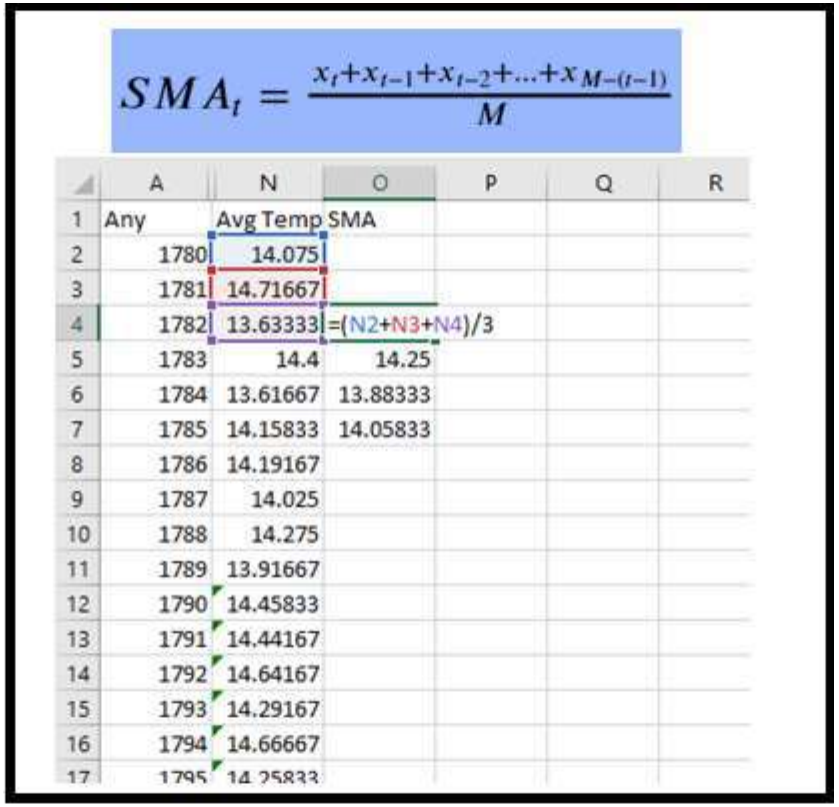
The commonly used time series method is the Moving Average. This method is slick with random short-term variations. Relatively associated with the components of time series.

- **The Moving Average (MA) (or) Rolling Mean:** The value of MA is calculated by taking average data of the time-series within k periods.
- Let’s see the types of moving averages:
- Simple Moving Average (SMA),
- Cumulative Moving Average (CMA)
- Exponential Moving Average (EMA)

Simple Moving Average (SMA)

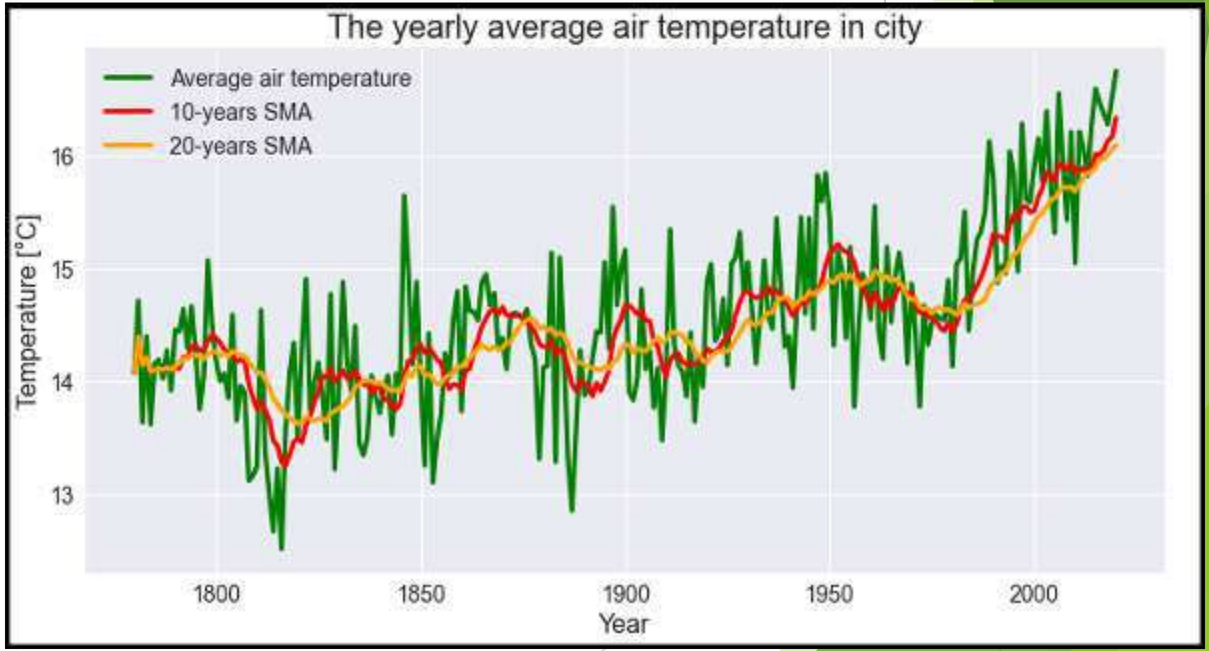
Simple Moving Average (SMA)

- The Simple Moving Average (SMA) calculates the unweighted mean of the previous M or N points. We prefer selecting sliding window data points based on the amount of smoothing, as increasing the value of M or N improves smoothing but reduces accuracy.



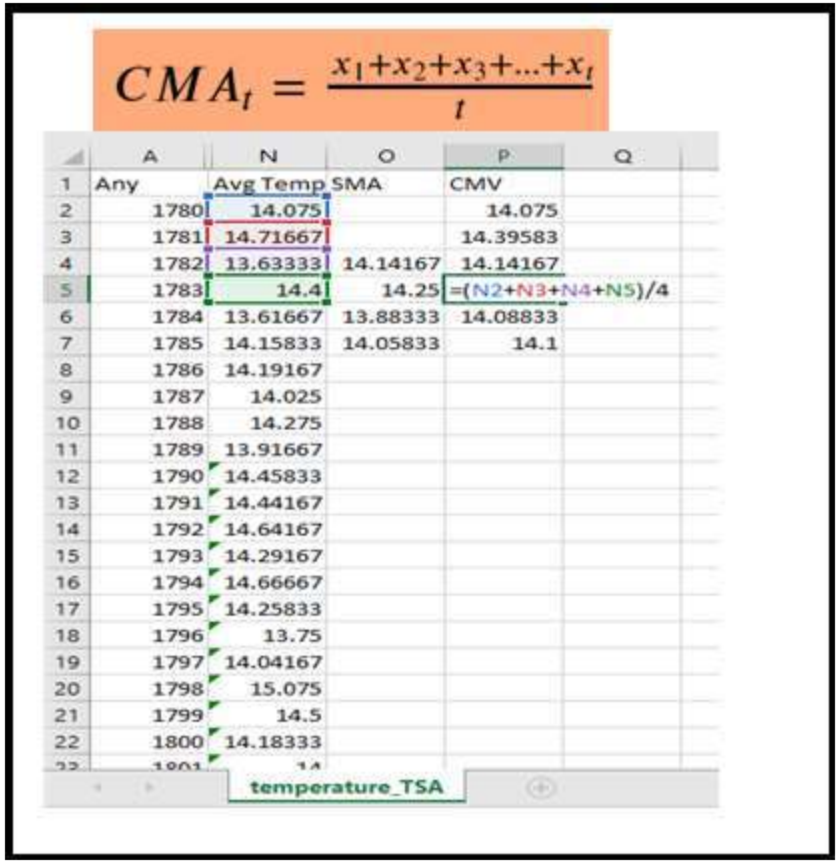
$$SMA_t = \frac{x_t + x_{t-1} + x_{t-2} + \ldots + x_{M-(t-1)}}{M}$$

	A	N	O	P	Q	R
1	Any	Avg Temp	SMA			
2	1780	14.075				
3	1781	14.71667				
4	1782	13.63333	=(N2+N3+N4)/3			
5	1783	14.4	14.25			
6	1784	13.61667	13.88333			
7	1785	14.15833	14.05833			
8	1786	14.19167				
9	1787	14.025				
10	1788	14.275				
11	1789	13.91667				
12	1790	14.45833				
13	1791	14.44167				
14	1792	14.64167				
15	1793	14.29167				
16	1794	14.66667				
17	1795	14.25833				



Cumulative Moving Average (CMA)

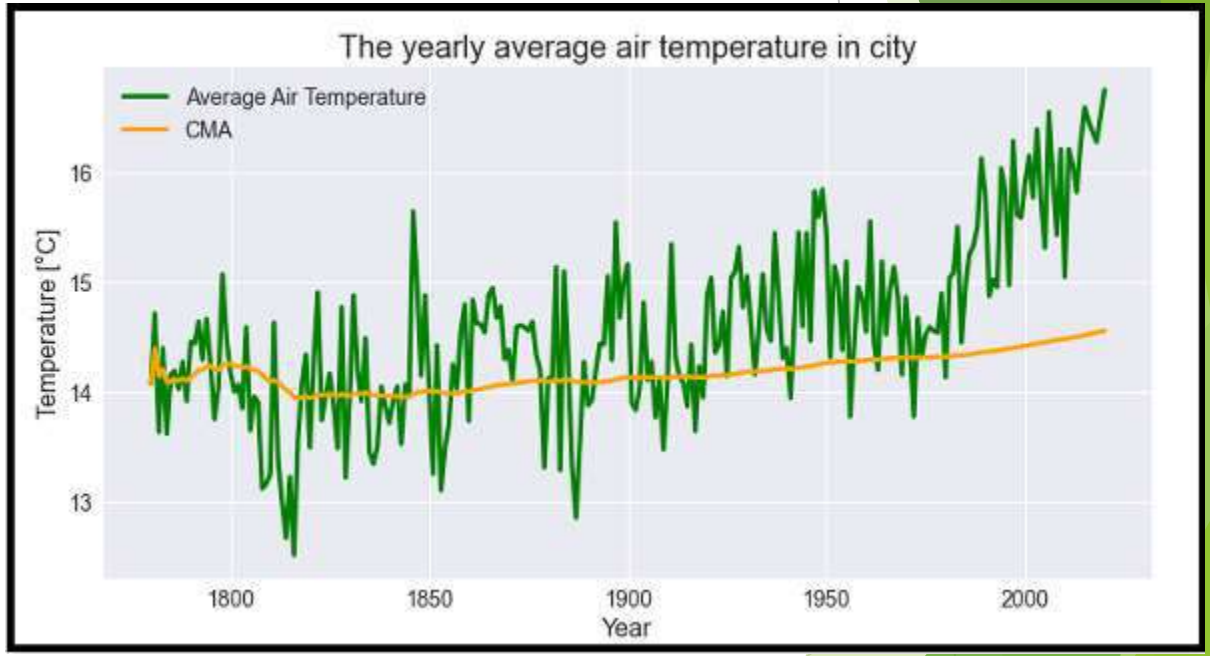
The CMA is the unweighted mean of past values till the current time.



$$CMA_t = \frac{x_1+x_2+x_3+...+x_t}{t}$$

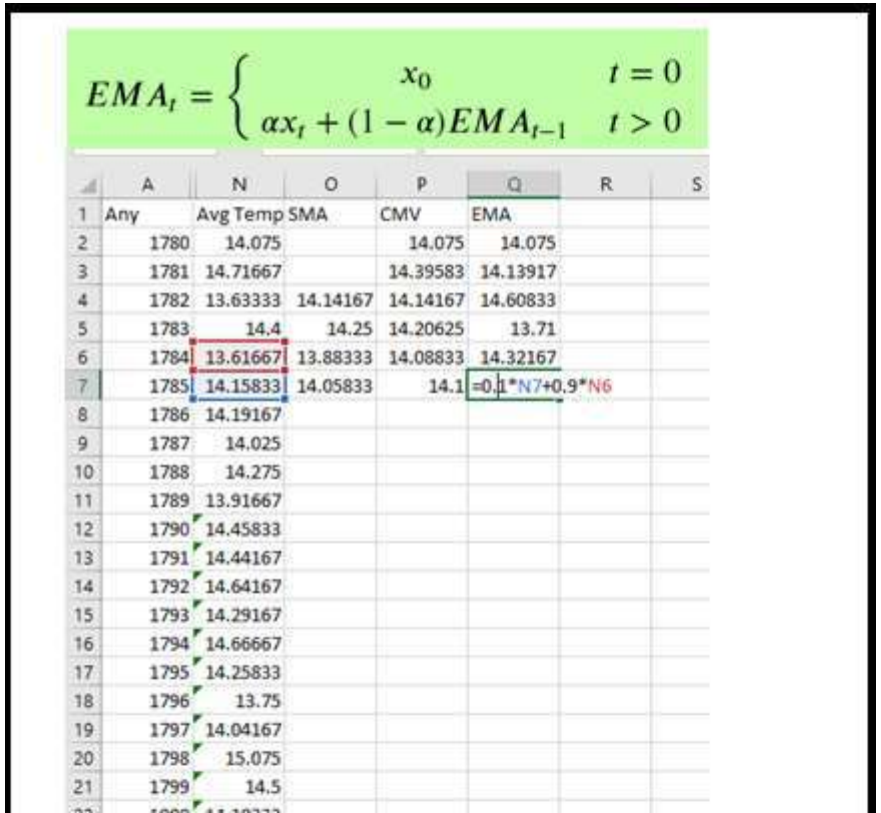
	A	N	O	P	Q
1	Any	Avg Temp	SMA	CMV	
2	1780	14.075		14.075	
3	1781	14.71667		14.39583	
4	1782	13.63333	14.14167	14.14167	
5	1783	14.4	14.25	=(N2+N3+N4+N5)/4	
6	1784	13.61667	13.88333	14.08833	
7	1785	14.15833	14.05833	14.1	
8	1786	14.19167			
9	1787	14.025			
10	1788	14.275			
11	1789	13.91667			
12	1790	14.45833			
13	1791	14.44167			
14	1792	14.64167			
15	1793	14.29167			
16	1794	14.66667			
17	1795	14.25833			
18	1796	13.75			
19	1797	14.04167			
20	1798	15.075			
21	1799	14.5			
22	1800	14.18333			
23	1801	14			

temperature_TSA



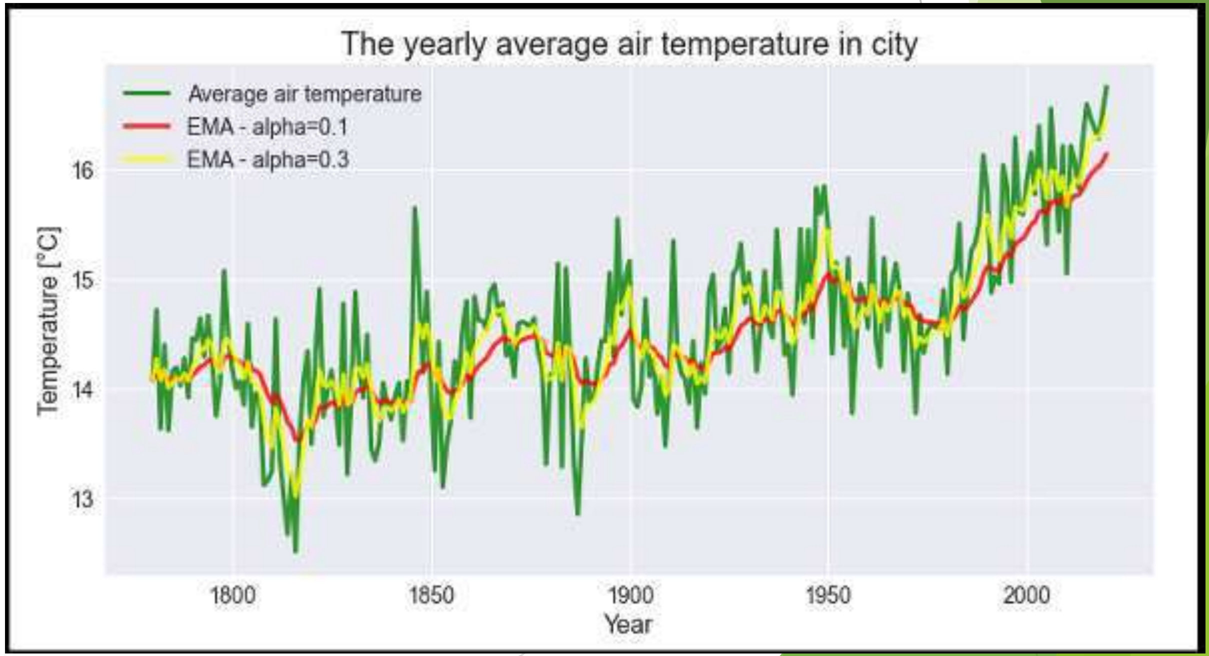
Exponential Moving Average (EMA)

- EMA is mainly used to identify trends and filter out noise. The weight of elements is decreased gradually over time. This means It gives weight to recent data points, not historical ones. Compared with SMA, the EMA is faster to change and more sensitive.
- α –>Smoothing Factor.
- It has a value between 0,1.
- Represents the weighting applied to the very recent period.



$$EMA_t = \begin{cases} x_0 & t = 0 \\ \alpha x_t + (1-\alpha)EMA_{t-1} & t > 0 \end{cases}$$

	A	N	O	P	Q	R	S
1	Any	Avg Temp	SMA	CMV	EMA		
2	1780	14.075		14.075	14.075		
3	1781	14.71667		14.39583	14.13917		
4	1782	13.63333	14.14167	14.14167	14.60833		
5	1783	14.4	14.25	14.20625	13.71		
6	1784	13.61667	13.88333	14.08833	14.32167		
7	1785	14.15833	14.05833	14.1	=0.1*N7+0.9*N6		
8	1786	14.19167					
9	1787	14.025					
10	1788	14.275					
11	1789	13.91667					
12	1790	14.45833					
13	1791	14.44167					
14	1792	14.64167					
15	1793	14.29167					
16	1794	14.66667					
17	1795	14.25833					
18	1796	13.75					
19	1797	14.04167					
20	1798	15.075					
21	1799	14.5					



ACF and PACF

- In time series analysis, ACF and PACF plots are two of the most important tools for identifying the underlying structure of a time series. The ACF plot shows the correlation of a time series with itself at different lags, while the PACF plot shows the correlation of a time series with itself at different lags, after removing the effects of the previous lags.

Autocorrelation Function (ACF)

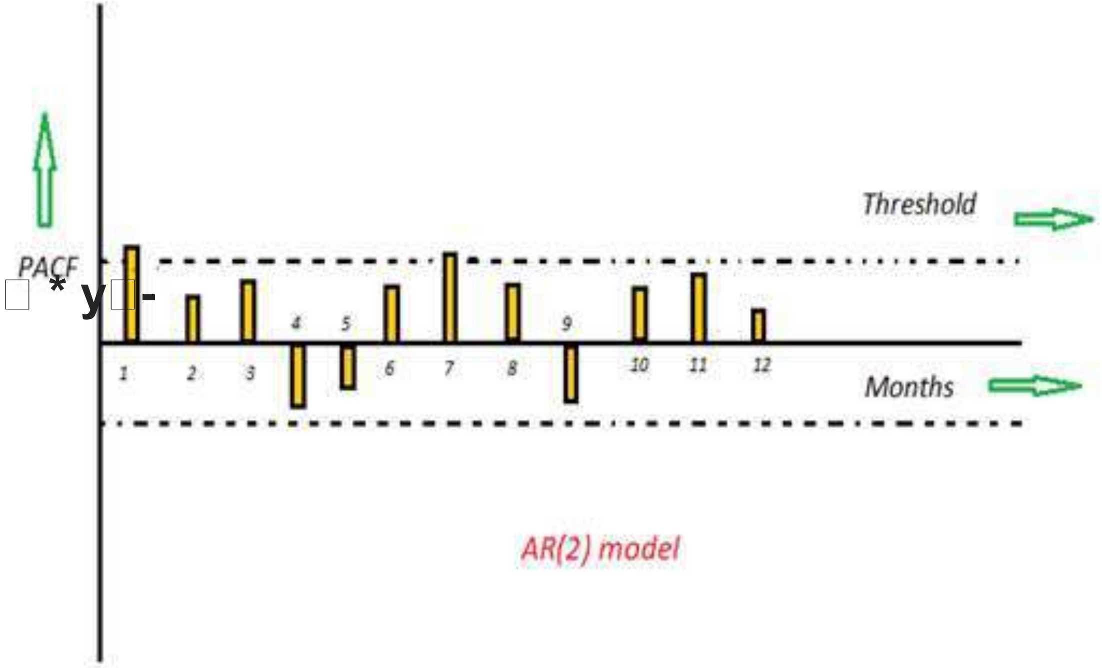
- The ACF plot is a graphical representation of the correlation of a time series with itself at different lags. The correlation coefficient is a measure of how closely two variables are related. A correlation coefficient of 1 indicates a perfect positive relationship, while a correlation coefficient of -1 indicates a perfect negative relationship. A correlation coefficient of 0 indicates no relationship between the two variables.
- The ACF plot can be used to identify the order of an AR model. The order of an AR model is the number of lags that are included in the model. The ACF plot will show spikes at the lags that are included in the model.

Partial Autocorrelation Function (PACF)

- The PACF plot is a graphical representation of the correlation of a time series with itself at different lags, after removing the effects of the previous lags. The PACF plot can be used to identify the order of an MA model. The order of an MA model is the number of lags that are included in the model. The PACF plot will show spikes at the lags that are included in the model.

AR (Auto-Regressive) Model

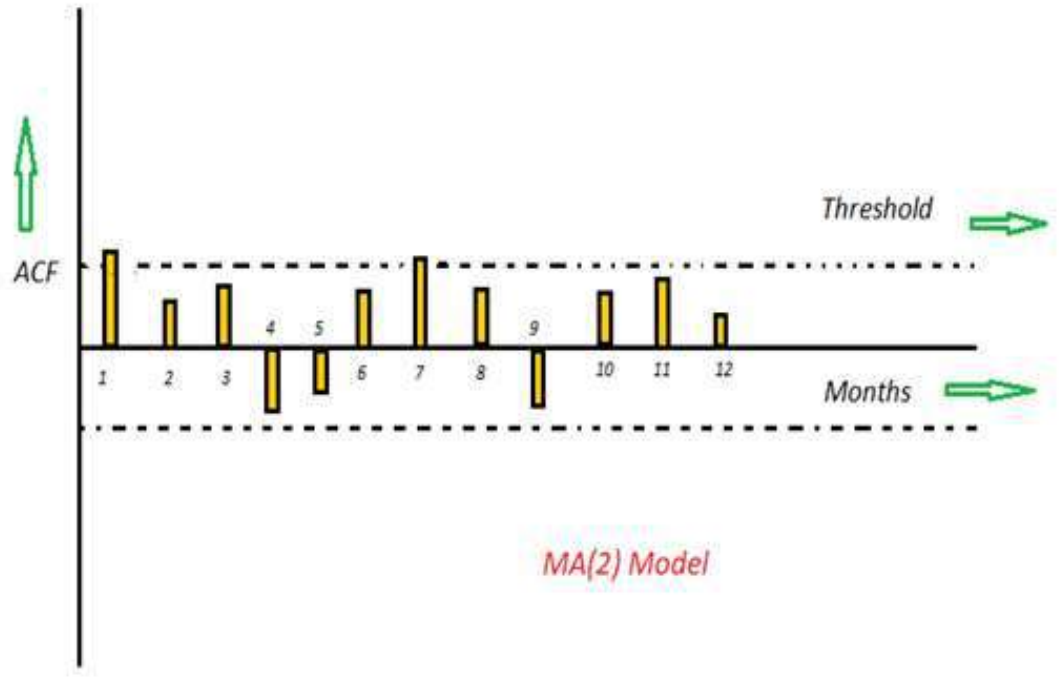
$$Yt = \beta_1 * y_{-1} + \beta_2 * y_{\square-2} + \beta_3 * y_{\square-3} + \ldots\ldots\ldots\ldots + \beta_{\square} * y_{\square-\square}$$



MA (Moving Average) Model

This kind of model calculates the residuals or errors of past time series and calculates the present or future values in the series in know as Moving Average (MA) model.

$Y_t = \alpha_1 * Ɛ_{t-1} + \alpha_2 * Ɛ_{t-2} + \alpha_3 * Ɛ_{t-3} + \ldots\ldots\ldots\ldots + \alpha_q * Ɛ_{t-q}$



ARMA (Auto Regressive Moving Average) Model

- This is a model that is combined from the AR and MA models. In this model, the impact of previous lags along with the residuals is considered for forecasting the future values of the time series. Here β represents the coefficients of the AR model and α represents the coefficients of the MA model.
- $\mathbf{Yt = \beta_1 * y_{\square-1} + \alpha_1 * \varepsilon_{\square-1} + \beta_2 * y_{\square-2} + \alpha_2 * \varepsilon_{\square-2} + \beta_3 * y_{\square-3} + \alpha_3 * \varepsilon_{\square-3} + \ldots\ldots\ldots\ldots + \beta_{\square} * y_{\square-\square} + \alpha_{\square} * \varepsilon_{\square-\square}}$

ARIMA (Auto-Regressive Integrated Moving Average) Model

- We know that in order to apply the various models we must in the beginning convert the series into Stationary Time Series. In order to achieve the same, we apply the differencing or Integrated method where we subtract the t-1 value from t values of time series. After applying the first differencing if we are still unable to get the Stationary time series then we again apply the second-order differencing.
- The ARIMA model is quite similar to the ARMA model other than the fact that it includes one more factor known as Integrated(I) i.e. differencing which stands for I in the ARIMA model. So in short ARIMA model is a combination of a number of differences already applied on the model in order to make it stationary, the number of previous lags along with residuals errors in order to forecast future values.

ARIMA is an acronym that stands for Auto-Regressive Integrated Moving Average.

Specifically, –

AR Autoregression.A model that uses the dependent relationship between an observation and some number of lagged observations.

I Integrated. The use of differencing of raw observations in order to make the time series stationary.

MA Moving Average. A model that uses the dependency between an observation and a residual error from a moving average model applied to lagged observations.

Each of these components are explicitly specified in the model as a parameter.

Note that AR and MA are two widely used linear models that work on stationary time series, and I is a preprocessing procedure to “stationarize” time series if needed.

A standard notation is used of ARIMA(p, d, q) where the parameters are substituted with integer values to quickly indicate the specific ARIMA model being used. –

p The number of lag observations included in the model, also called the lag order. –

d The number of times that the raw observations are differenced, also called the degree of differencing. –

q The size of the moving average window, also called the order of moving average.